

CallForge — User Manual

A targeted-capture germline analysis pipeline (FASTQ → annotated joint callset → burden)

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1. Introduction

CallForge is a disease- and organism-agnostic **Nextflow DSL2** pipeline that turns demultiplexed paired-end FASTQs from a hybridization-capture panel into a high-quality, joint-genotyped, **annotated** callset spanning **SNV/indel**, **CNV**, **STR** and **paralog-aware** variants, and carries through to a **rare-variant burden / association** layer.

CallForge is the **analysis counterpart to CaptureForge** (the probe-design pipeline) and **consumes CaptureForge’s handoff**: the as-built target `final_covered_targets.bed` plus per-gene metadata (design class, CNV callability, paralog flags, STR loci, burden groups). That design-time knowledge drives and labels the matching analysis steps — the **CaptureForge → CallForge interlock**.

Its immediate use is a 59-gene human malaria-susceptibility panel on African-ancestry dried-blood-spot (DBS) samples, but it generalizes to **any panel, cohort, and (where feasible) organism** by changing inputs and configuration only — no code edits.

Who it is for. Bioinformaticians and genomics-core staff running targeted germline analysis who need a reproducible, QC-gated workflow with honest reporting. The pipeline emphasizes **QC at every stage with captioned plots, fail-loud invariants** (reference and annotation-database scope), and a **flag-and-quarantine** QC gate that never silently drops data.

Status note used throughout this manual. CallForge is implemented and verified end-to-end on a small synthetic **test** profile. Where a number is a property of that synthetic data (not of a real cohort), the text and figure captions say so explicitly — *machinery-proven, not data quality*. See §11 (Interpreting results) and §12 (Real-run checklist).

2. Concepts

Targeted-capture germline analysis. A hybridization-capture panel enriches sequencing for a defined set of target regions (here, CaptureForge’s `final_covered_targets.bed`). Reads are aligned to a reference, duplicates marked, and variants called *jointly* across the cohort so that genotypes — including homozygous-reference calls — are consistent and comparable across samples.

Joint genotyping. Per-sample gVCFs (GATK HaplotypeCaller) are combined (GenomicsDBImport or CombineGVCFs) and genotyped together (GenotypeGVCFs). Joint genotyping yields a square cohort matrix and well-calibrated genotypes, which matter for downstream cohort QC and burden testing.

The four variant classes (why each needs its own caller).

Class	Caller	Why a dedicated method
SNV / indel	GATK HaplotypeCaller (or DeepVariant)	local re-assembly of haplotypes
CNV	CNVkit (or GATK gCNV)	read-depth ratios vs a panel-of-normals, not per-base genotypes
STR	ExpansionHunter	repeat-aware genotyping from spanning/flanking reads
paralog-aware	<code>paralog_flag.py</code> confidence layer	multi-mapping ambiguity is a <i>confidence</i> problem, not a new caller

The CaptureForge → CallForge interlock. CaptureForge records, per gene, what it could and could not design well. CallForge consumes that:

- **CNV callability** — depth-callable genes (e.g. GYPC, HP) vs breakpoint-blind genes (e.g. CR1, CFH) label every CNV call’s confidence.
- **STR catalog** — STR-class targets seed the ExpansionHunter variant catalog.
- **Paralog flags** — paralog-ambiguous genes (e.g. CD209, CASP1, CR1, HP, CFH) drive the paralog confidence layer.
- **Burden groups** — collapse units for gene-set burden testing.

The reference invariant. CallForge aligns to the **same no-alt GRCh38 primary assembly CaptureForge designed against**, and asserts it fail-loud: no alt/decoy contigs, and the target BED’s build must match the reference (see §9, §13).

Rare-variant burden testing. Individually rare variants are aggregated per gene (and per gene-set) and tested for association with phenotype, adjusting for covariates and ancestry. CallForge filters to **rare + functional** variants, collapses by **gene** and **CaptureForge burden group**, and runs a gene-burden test (§7, §10).

3. Pipeline architecture

CallForge runs **16 stages (0–15)**. Each stage writes metrics (JSON/TSV), emits captioned plots (PNG+SVG), and feeds MultiQC plus a self-contained cohort QC dashboard. The diagram below is the authoritative map; external inputs (dashed = drives/labels downstream stages) feed in from the side.

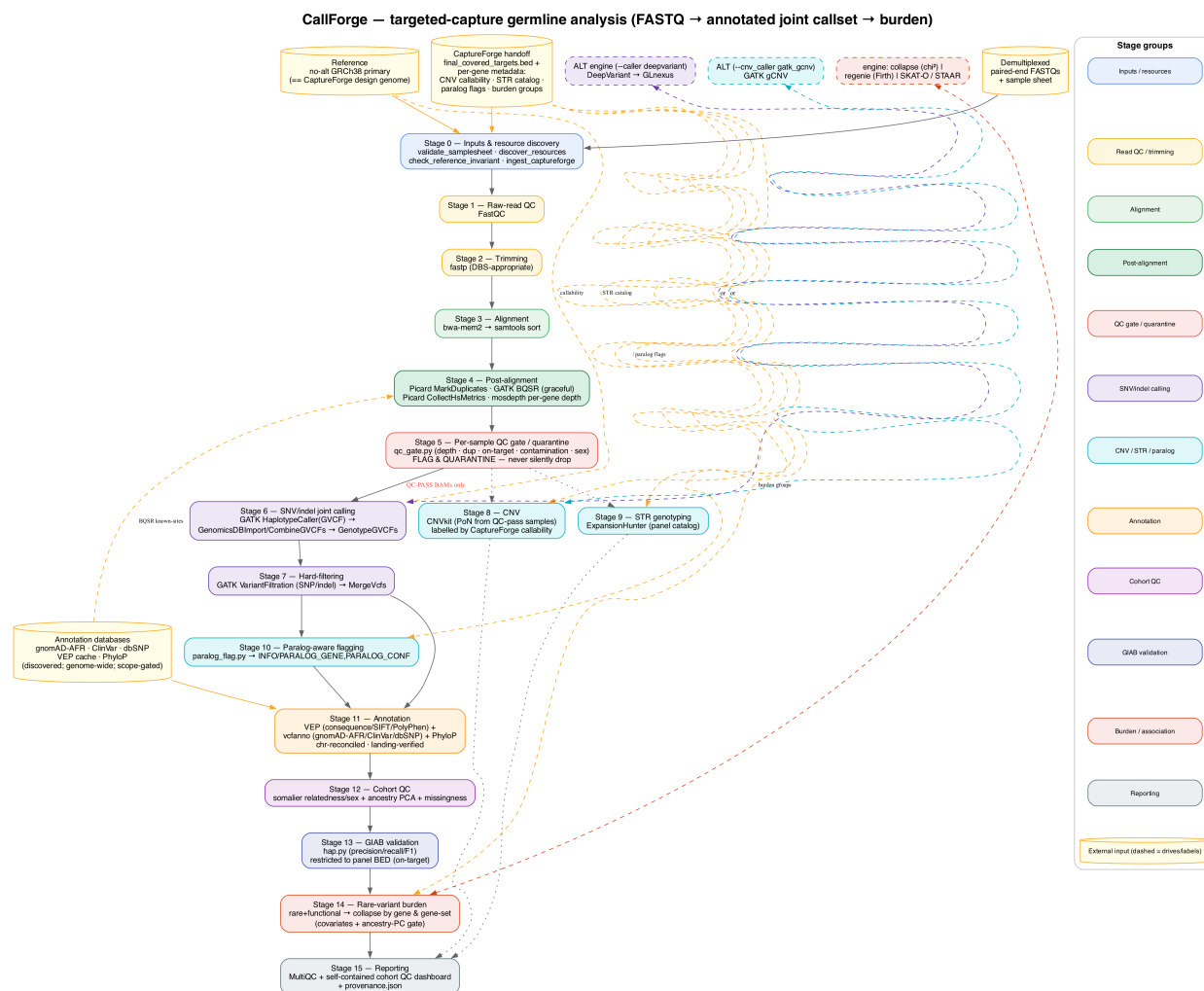


Figure 1: CallForge pipeline DAG — stages colour-coded by group; external inputs (reference, CaptureForge handoff, annotation databases, FASTQs) feed in; alternative engines (DeepVariant+GLnexus, gatk_gcnv, collapse/regenie/SKAT-O) shown as dashed branches.

Stage walkthrough (real inputs → outputs).

#	Stage	Key tool(s)	Output dir + key file(s)
0	Inputs & resource discovery	validate_samplesheet, discover_resources, ingest_captureforge	stage0_inputs/ — manifest, invariant, gene_metadata.tsv

#	Stage	Key tool(s)	Output dir + key file(s)
1	Raw-read QC	FastQC	stage1_rawqc/
2	Trimming	fastp	stage2_trim/
3	Alignment	bwa-mem2, samtools sort	stage3_align/
4	Post-alignment	Picard MarkDuplicates, GATK BQSR, CollectHsMetrics, mosdepth	stage4_postalign/ — HsMetrics, mosdepth per-gene depth
5	Per-sample QC gate	qc_gate.py	stage5_qc_gate/ — qc_pass.txt, qc_quarantine.txt
6	SNV/indel joint calling	GATK HaplotypeCaller, GenomicsDBImport, GenotypeGVCFs	stage6_calling/ — joint.vcf.gz
7	Hard-filtering	GATK VariantFiltration, MergeVcfs	stage7_filter/ — joint.filtered.vcf.gz
8	CNV	CNVkit	stage8_cnv/ — cnv_calls.tsv (with callability)
9	STR	ExpansionHunter	stage9_str/ — str_calls.tsv
10	Paralog-aware	paralog_flag.py	stage10_paralog/ — paralog.annotated.vcf.gz
11	Annotation	VEP, vcfanno, PhyloP	stage11_annotation/ — annotated.vcf.gz, variants.flat.tsv
12	Cohort QC	somalier	stage12_cohortqc/
13	GIAB validation	hap.py	stage13_giab/
14	Burden / association	collapse / regenie / SKAT-O	stage14_burden/ — burden_results.tsv
15	Reporting	MultiQC, dashboard, provenance	stage15_report/

Nextflow / profile model. `main.nf` calls one subworkflow (`subworkflows/callforge.nf`) that wires modules in `modules/`. Behaviour is set by **profiles** composed on the command line: an execution profile (`test`, `mac_local`, `hpc_slurm`) plus an environment engine (`conda`, `docker`, `apptainer`). Each process declares its own pinned conda env (`env/*.yaml`) or a pinned container (VEP, hap.py), so `-profile ... ,conda` or `,apptainer` builds exactly what each stage needs.

Engine options (see §7 for full defaults): SNV/indel `--caller gatk|deepvariant`; joint method `--joint_method genomicsdb|combinegvcfs`; CNV `--cnv_caller cnvkit|gatk_gcnv`; CNVkit segmentation `--cnv_segment_method cbs|haar|hmm`; burden `--burden_engine regenie|skat|collapse`.

4. Installation

CallForge needs **Nextflow** (≥ 23.10) and either conda/mamba or a container engine. Each stage runs in its own pinned environment (`env/*.yaml`) or pinned container, so you do not install tools by hand — `-profile conda` (or `,apptainer`) builds each environment on first use.

4.1 Local install (personal computer)

Common steps.

```
# 1. Nextflow (Java 17+ required)
curl -s https://get.nextflow.io | bash && sudo mv nextflow /usr/local/bin/
# or: mamba create -n nextflow -c bioconda 'nextflow>=23.10'
```

```
# 2. obtain CallForge and enter the repository
cd /path/to/callforge
```

```
# 3. core environment (others build on demand)
mamba env create -f env/callforge.yaml
```

macOS (Apple Silicon). Some bioconda tools lack `osx-arm64` builds. Two options:

```
# (a) Rosetta-emulated x86 env (documented path; the report flags emulated
↪ stages)
CONDA_SUBDIR=osx-64 mamba env create -f env/callforge.yaml
```

```
# (b) native arm64 - works for most tools, but pin mosdepth >= 0.3.11
#      (mosdepth 0.3.8 has no arm64 build; 0.3.8 + Rosetta is the alternative)
```

VEP and hap.py are run from their **official containers** (robust across platforms); ensure Docker is available and use `-profile <exec>,docker`.

Linux. Native; `mamba env create -f env/*.yaml` resolves cleanly, or use the container route.

Container route. Build the core image once and run containerized:

```
docker build -t callforge:0.1.0 -f env/Dockerfile .
# Apptainer (HPC): apptainer build callforge.sif env/callforge.def
```

4.2 Shared HPC install (Ubuntu 24.04 SLURM cluster)

The production cohort run is intended for the cluster (native linux-64; no Rosetta). The `hpc_slurm` profile uses the SLURM executor (global, cluster-wide across the nodes) and Apptainer for the containerized tools.

```
# shared conda at /hpc/opt/conda
source /hpc/opt/conda/etc/profile.d/conda.sh
mamba env create -p /hpc/opt/conda/envs/callforge -f env/callforge.yaml
mamba env create -p /hpc/opt/conda/envs/callforge-annotate -f env/annotate.yaml
# ... and cohortqc / cnv / str / happy / burden / deepvariant as needed
```



```
# Apptainer image for the core tools (Docker is usually forbidden on clusters)
apptainer build callforge.sif env/callforge.def
export APPTAINER_BINDPATH=/data/refs,/data/annotation_db,$HOME/.vep    # mount
↳ resources
```

```
# run
nextflow run main.nf -profile hpc_slurm,apptainer -params-file params.full.yaml \
  --slurm_partition compute \
  --slurm_account mylab \
  --scratch_dir /scratch/$USER
```

An **Lmod** module + launcher wrapper is the recommended front door (set PATH to the shared Nextflow, source the shared conda, set APPTAINER_BINDPATH); see docs/hpc_deployment.md. conf/hpc_slurm.config sets a shared conda cache so each env builds once cluster-wide; --slurm_partition / --slurm_account / --scratch_dir are overridable. Heavy steps (index, align, large labels) get bigger slots.

5. Inputs

5.1 Sample sheet (CSV) — `--input`

Columns: `sample_id`, `fastq_1`, `fastq_2`, `sex`, `phenotype`, `covariate_*`, `batch`. Example:

```
sample_id,fastq_1,fastq_2,sex,phenotype,covariate_age,batch
S001,/data/fq/S001_R1.fastq.gz,/data/fq/S001_R2.fastq.gz,F,case,41,b1
S002,/data/fq/S002_R1.fastq.gz,/data/fq/S002_R2.fastq.gz,M,control,55,b1
HG002,/data/fq/HG002_R1.fastq.gz,/data/fq/HG002_R2.fastq.gz,M,control,30,b2
```

- `sample_id` unique; `fastq_1`/`fastq_2` are the paired-end reads (validated to exist and differ). `sex` is M/F/U (free tokens normalized).
- **phenotype is required only for the burden layer** (case/control or quantitative). Without it, the pipeline runs through the annotated callset and **skips burden with a message** (`burden_meta.json` records `status: skipped_no_phenotype`).
- `covariate_*` columns (any number) become burden covariates; `batch` is carried for QC.

Validation (`validate_samplesheet.py`) writes `samplesheet.valid.csv` and `samplesheet_summary.json` (with `burden_eligible`).

5.2 CaptureForge handoff — `--target_bed` (+ metadata)

Point CallForge at CaptureForge’s as-built outputs:

```
target_bed:
  ↪ /path/captureforge/results/full_run/stage9_qc/final_covered_targets.bed
captureforge_dir: /path/captureforge/results/full_run # auto-finds metrics.json
  ↪ + baits.csv
paralog_genes: "CD209,CASP1,CR1,HP,CFH"
```

`final_covered_targets.bed` uses `GENE|class` names (e.g. `ATP2B4|coding`, `GYPC|cnv`). `metrics.json` supplies CNV callability; `baits.csv` an off-target signal. If CaptureForge outputs are absent, supply `--gene_metadata` (a per-gene TSV), `--paralog_genes`, and `--str_catalog` directly. `ingest_captureforge.py` produces `gene_metadata.tsv` with columns `gene`, `classes`, `...`, `is_cnv_target`, `cnv_callable`, `cnv_callable_reason`, `is_str_target`, `str_loci`, `is_paralog`, `off_target_frac`, `low_coverage`, `burden_group`.

5.3 Reference — `--genome_fasta`

The **no-alt GRCh38 primary assembly** the panel was designed against (Ensembl-named, e.g. `1,2,...,22,X,Y,MT`). `--genome_build GRCh38`. CallForge builds `.fai/.dict/bwa-mem2` index if absent and asserts the reference invariant (§9).

6. Annotation databases

Annotation is the most resource-sensitive stage and is treated carefully: CallForge **discovers** local resources, **validates their genomic scope** (fail-loud on region-limited databases), and **reconciles contig naming** so values are never silently missed. Nothing is downloaded unless a resource is missing **and** required **and** you opt in (`--allow_download`).

6.1 The databases

Database	Provides	Stage(s) / param	Scope
gnomAD (AFR)	population allele frequencies, incl. African-ancestry AF_afr	11 (vcfanno), 14 (rare filter); -gnomad_vcf, -gnomad_af_field	genome-wide; scope-gated
ClinVar	clinical significance (CLNSIG, CLNDN, CLNREVSTAT, ...)	11; -clinvar_vcf	genome-wide; scope-gated
dbSNP	rsIDs (ID column)	11 (rsID), 4 (BQSR); -dbSNP_vcf, -known_sites	genome-wide; scope-gated
VEP cache	consequence, transcript, SIFT/PolyPhen, canonical/MANE	11; -vep_cache, -vep_release	species + assembly match
PhyloP	conservation score (bigWig)	11; -phyloP_bw	genome-wide

Namespacing. gnomAD INFO fields are written back as `gnomad_*` (e.g. `gnomad_AF_afr`) so the cohort's own AF/AC/AN are **never overwritten** — the classic annotation enrichment bug is designed out.

6.2 The genomic-scope gate (fail-loud)

A region-subset database (e.g. a single-locus gnomAD) silently returns “no AF / no ClinVar” outside its region — which a downstream filter would misread as *rare-novel*, corrupting burden testing. `discover_resources.py` therefore validates, per annotation VCF, that its data **spans the target panel's contigs** (tabix-probed, chr/no-chr reconciled), checks the **BGZF-EOF marker** (rejects truncated downloads), and **fails loud** on a found-but-unfit database:

```
[discover_resources] UNFIT (found but region-limited): gnomad_vcf: data spans
↳ only
1/19 target contigs (frac=0.0526 < 0.9); region-subset DB - replace with
↳ genome-wide.
[discover_resources] FAIL: ... Refusing to under-annotate.
```

Tune with `--scope_min` (default 0.9); override only with `--ignore_unfit_resources` (not advised) or `--allow_download`.

6.3 Contig-naming / chr-reconciliation

The design reference / callset is **Ensembl-numeric** (1), gnomAD and PhyloP are usually **chr-prefixed** (chr1), and NCBI ClinVar is Ensembl-numeric. CallForge reconciles naming **per resource**: the query is renamed to a resource's style only where it differs, then renamed back — and the annotation step then **verifies the values actually landed** (verify_annotation.py, fail-loud), so a contig-name mismatch can never silently drop an annotation.

6.4 Fetching databases — bin/fetch_annotation_dbs.sh

A resumable downloader for genome-wide gnomAD v4.1 + ClinVar (GRCh38):

```
# panel-slice mode (recommended for a targeted panel; a few hundred MB)
MODE=panel \
PANEL_BED=/path/final_covered_targets.bed \
OUTDIR=/path/annotation_db \
  bash bin/fetch_annotation_dbs.sh
```

```
# full genome-wide mode (very large; use on HPC with a fast network)
MODE=full DATASETS="genomes" bash bin/fetch_annotation_dbs.sh
```

Panel mode merges nearby intervals to cut HTTP round-trips, slices each chromosome with per-chromosome retry, and is resumable (re-run to continue, not restart). **dbSNP is not fetched by this script** — supply a genome-wide dbSNP separately (it is also a BQSR known-site).

6.5 Resource auto-discovery

discover_resources.py scans --resource_dirs (default \$HOME,/usr/local/share,/opt), validates each resource (build, index, scope, integrity), and writes stage0_inputs/resource_manifest.json. Missing human-only resources are reported and the dependent step skips gracefully. Explicit --vep_cache/--gnomad_vcf/--clinvar_vcf/... overrides take precedence and are still validated.

6.6 Bring-your-own / non-human databases

CallForge is organism-generic: VEP runs by --species (point at that species' VEP cache), and vcfanno reads a **bring-your-own-databases TOML** (--vcfanno_toml). Human-only resources (gnomAD/ClinVar/BQSR known-sites) **skip gracefully when absent**. To register a new organism: set --species, --genome_fasta (that build), --vep_cache, and a --vcfanno_toml listing whatever frequency/clinical databases exist; leave the human DB params unset.

7. Configuration

Set parameters via `-params-file params.yaml` (see `params.example.yaml`) or `--key value`. Profiles set executor + resource ceilings: **test** (3–4 synthetic samples, minutes), **mac_local** (full, one machine), **hpc_slurm** (SLURM + Apptainer/shared-conda).

Key tunables (real defaults).

Param	Default	Meaning
<code>--caller</code>	<code>gatk</code>	SNV/indel engine: <code>gatk</code> or <code>deepvariant</code>
<code>--joint_method</code>	<code>genomicsdb</code>	<code>genomicsdb</code> or <code>combinegvcfs</code>
<code>--ploidy</code>	<code>2</code>	HaplotypeCaller ploidy
<code>--cnv_caller</code>	<code>cnvkit</code>	<code>cnvkit</code> or <code>gatk_gcnv</code>
<code>--cnv_method</code>	<code>hybrid</code>	CNVkit hybrid (capture) or <code>amplicon</code>
<code>--cnv_segment_method</code>	<code>cbs</code>	<code>cbs</code> (R/DNAcopy), <code>haar</code> (pure-python), <code>hmm</code>
<code>--min_mean_target_depth</code>	<code>30</code>	QC gate: min mean target depth ($\times$)
<code>--max_dup_rate</code>	<code>0.40</code>	QC gate: max duplication fraction
<code>--min_on_target</code>	<code>0.40</code>	QC gate: min on-target fraction
<code>--max_contamination</code>	<code>0.03</code>	QC gate: max VerifyBamID2 FREEMIX
<code>--enforce_sex_check</code>	<code>true</code>	quarantine on sex mismatch (when assessable)
<code>--snp_filter_expr</code>	GATK WGS/WES default	SNP hard-filter JEXL
<code>--indel_filter_expr</code>	GATK WGS/WES default	indel hard-filter JEXL
<code>--scope_min</code>	<code>0.9</code>	min fraction of target contigs a DB must span
<code>--known_sites</code>	<code>null</code>	BQSR known-sites (dbSNP + Mills + 1000G)
<code>--somalier_sites</code>	<code>null</code>	somalier sites VCF (1000G for the build)
<code>--str_catalog</code>	<code>null</code>	ExpansionHunter catalog (else built from STR targets)
<code>--burden_engine</code>	<code>regenie</code>	<code>regenie</code> , <code>skat</code> , or <code>collapse</code>
<code>--burden_af_max</code>	<code>0.01</code>	rare threshold on gnomAD-AFR AF
<code>--burden_csq</code>	LoF + missense + splice + inframe set	qualifying consequences
<code>--n_ancestry_pcs</code>	<code>4</code>	ancestry PCs offered as covariates (PC-stability gate applies)

Param	Default	Meaning
<code>--vep_image</code>	ensembl-vep 110	VEP container (tag below)
<code>--happy_image</code>	hap.py 0.3.12	hap.py container (tag below)

The container defaults are `ensemblorg/ensembl-vep:release_110.0` (VEP) and `jmcdani20/hap.py:v0.3.12` (hap.py).

Hard-filter thresholds are the GATK WGS/WES defaults. Revisit them on real targeted data (panel Ti/Tv, pass rates, and QUAL/DP distributions guide tuning).

8. Running — step-by-step scenarios

All multi-flag commands use backslash continuation (one flag per line). On a local machine use `,conda` (and `,docker` for the VEP/hap.py containers); on the cluster use `,apptainer`.

8.1 Quickstart (test profile)

```
bash test/make_test_data.sh
nextflow run main.nf -profile test,docker
```

This builds a tiny self-contained fixture (mini-genome + synthetic reads) and runs all 16 stages in minutes. Use it to confirm your install.

8.2 Full human cohort (primary scenario)

```
# 1. write the sample sheet ($5.1) with phenotype if you want burden

# 2. confirm resources discover + pass the scope gate; fetch genome-wide
#   gnomAD/ClinVar if missing ($6.4); supply a genome-wide dbSNP

# 3. run (authoring machine)
nextflow run main.nf \
  -profile mac_local,conda \
  -params-file params.full.yaml

# 3'. or the cluster
nextflow run main.nf \
  -profile hpc_slurm,apptainer \
  -params-file params.full.yaml
```

`params.full.yaml` wires the reference, the CaptureForge handoff, resource directories, and engine choices; set `input:` to your cohort sample sheet. Inspect `results/stage5_qc_gate/qc_scorecard.tsv`, then the annotated callset and the dashboard.

8.3 Callset-only (no phenotype)

Omit `phenotype` (or set `--run_burden false`). The pipeline runs through Stage 11 (annotated callset) + cohort QC + GIAB and **skips burden**; `burden_meta.json` records the skip. Everything else (annotated VCF, per-variant TSV, CNV/STR/paralog) is produced.

8.4 Burden / association run (with phenotype)

```
nextflow run main.nf \
  -profile hpc_slurm,apptainer \
  -params-file params.full.yaml \
```

```

--run_burden true \
--burden_engine regenie \
--burden_af_max 0.01 \
--n_ancestry_pcs 4 \
--covariates covariate_age,covariate_sex

```

Burden filters to rare+functional variants, collapses by gene and CaptureForge burden group, and tests with covariates + ancestry PCs. The **ancestry-PC stability gate** drops the PCs and runs unadjusted (recorded in provenance) if there are too few PCA sites for the cohort — read `burden_results.tsv` (its `engine` column states the engine) with the QQ / Manhattan plots and the validity caveats (§11).

8.5 A different panel

Swap the CaptureForge handoff — no code change:

```

nextflow run main.nf -profile mac_local,conda \
--input samplesheet.csv \
--genome_fasta /ref/GRCh38_noalt.fa \
--target_bed /panelB/final_covered_targets.bed \
--captureforge_dir /panelB/results/full_run

```

8.6 A different organism

```

nextflow run main.nf -profile mac_local,conda \
--input samplesheet.csv \
--genome_build <BUILD> \
--genome_fasta /ref/<species>_noalt.fa \
--target_bed /panel/final_covered_targets.bed \
--species <vep_species> \
--vep_cache /vep/<species> \
--vcfanno_toml /db/<species>.toml \
--run_burden false

```

Human-only resources (gnomAD/ClinVar/BQSR) skip gracefully. Minimum inputs: sample sheet, no-alt reference for that build, target BED, a VEP cache (or `--vep_mode gtf` with a `--gtf`), and a vcfanno TOML for any frequency/clinical databases you have.

8.7 HPC run

See §4.2. `-profile hpc_slurm,apptainer`; SLURM is global/cluster-wide; resources are copied to a shared path and bound into Apptainer.

8.8 Choosing engine alternatives

```

--caller deepvariant           # DeepVariant + GLnexus (container) instead of
↪ GATK
--cnv_caller gatk_gcncv       # GATK gCNV instead of CNVkit

```



```
--burden_engine skat           # SKAT-O / STAAR (R) instead of regenie
--burden_engine collapse      # chi-square collapsing (small-N /
↪ dependency-free)
```

Validate before production. DeepVariant+GLnexus, regenie, and SKAT-O/STAAR are wired but have **not yet been executed end-to-end** (the synthetic test cohort is too small). Exercise them on a real/realistic cohort before relying on them. The tested engines are GATK/GenomicsDB (calling) and collapse (burden).

8.9 GIAB validation run

Add a GIAB control to the sample sheet and supply truth:

```
nextflow run main.nf -profile mac_local,conda -params-file params.full.yaml \
  --giab_control_id HG002 \
  --giab_truth_vcf /giab/HG002_v4.2.1_benchmark.vcf.gz \
  --giab_truth_bed /giab/HG002_v4.2.1_benchmark.bed
```

hap.py scores precision/recall/F1 **restricted to the panel BED** (on-target only), so off-target truth cannot distort the panel's measured sensitivity.

9. Quality control

CallForge’s QC philosophy: **a plot at every applicable stage** (each with a one-line caption), gates that **flag and quarantine** rather than silently drop, and a single cohort QC dashboard plus MultiQC. The figures below are from the synthetic **test** run — they prove the machinery; real numbers come from a real cohort.

9.1 Read & alignment QC (Stages 1–4)

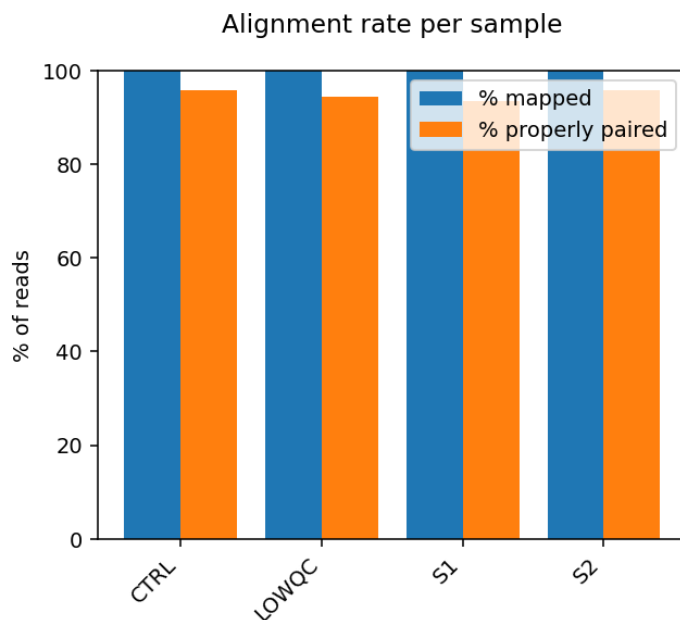


Figure 2: Percent of reads mapped and properly paired per sample (alignment success).

`CollectHsMetrics` (vs the target BED) gives on-target %, fold-enrichment, % target $\geq 10/20/30\times$, and uniformity (FOLD_80); `mosdepth` gives per-target and **per-gene depth** in the CaptureForge track style. Duplicates are marked by coordinate (correct for capture). BQSR runs when known-sites are available and **skips gracefully** otherwise.

9.2 Per-sample QC gate & quarantine (Stage 5)

`qc_gate.py` aggregates the per-sample metrics and applies thresholds (depth, dup, on-target, contamination, sex). Failing samples are **flagged and QUARANTINED** — written to `qc_quarantine.txt` with reasons, **excluded from joint calling and from burden counts**, but never silently dropped. Metrics that cannot be assessed (contamination without a VerifyBamID2 SVD panel; sex on an autosomal panel with no X/Y targets) are recorded **NA**, not failed.

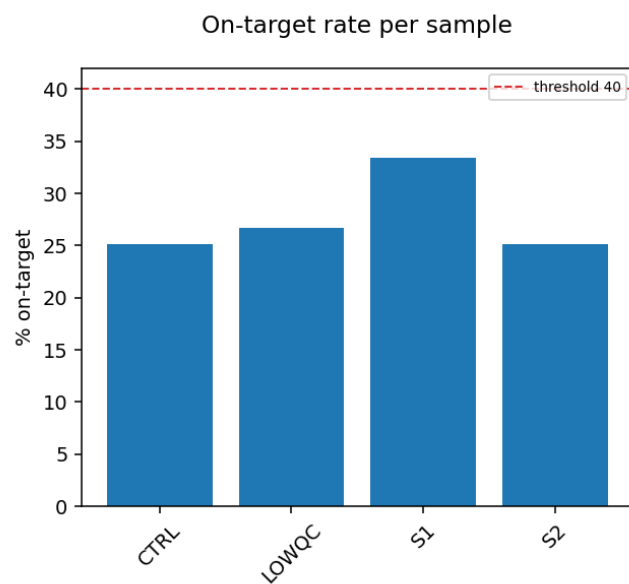


Figure 3: Percent of bases on the capture target per sample (capture specificity).

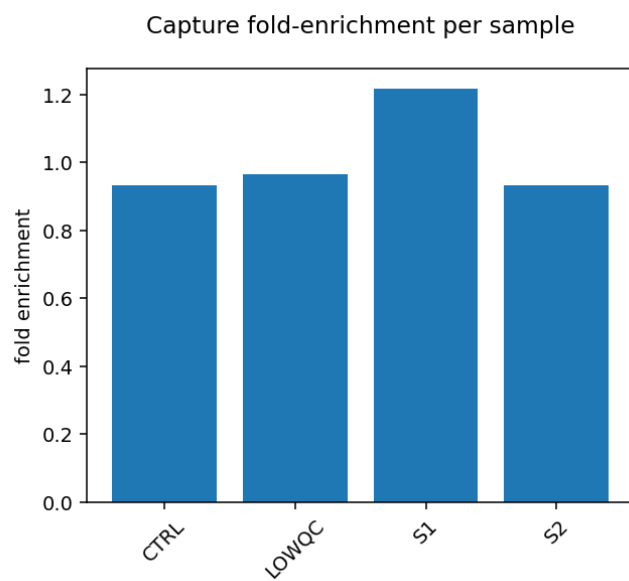


Figure 4: Fold-enrichment of on-target vs genome-average coverage (capture performance).

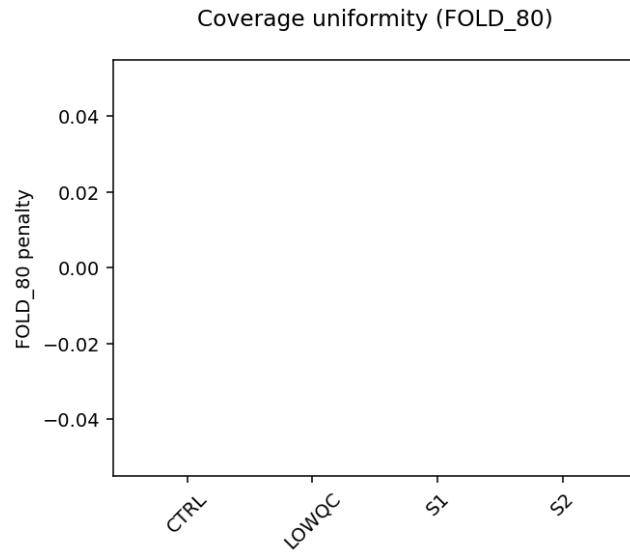


Figure 5: FOLD_80_BASE_PENALTY: fold extra sequencing for 80% of targets to reach the mean (1.0 = perfectly uniform).

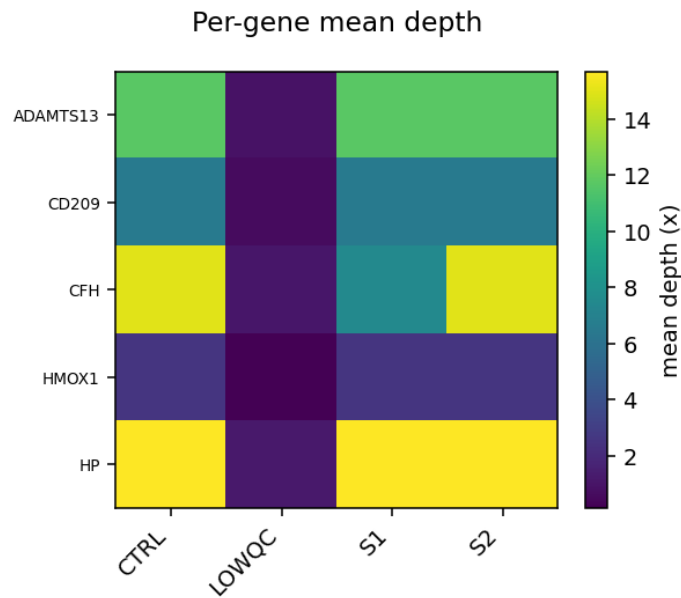


Figure 6: Mean coverage depth per target gene per sample; spot under-covered genes (e.g. CR1/CFH).

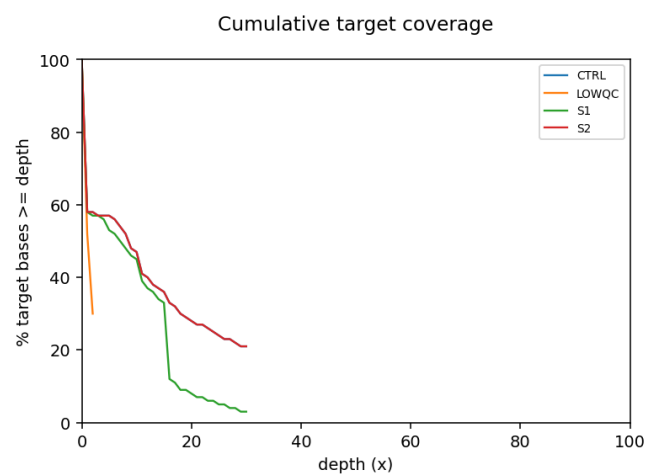


Figure 7: Fraction of target bases covered at $\geq$ each depth, per sample (sensitivity for calling).

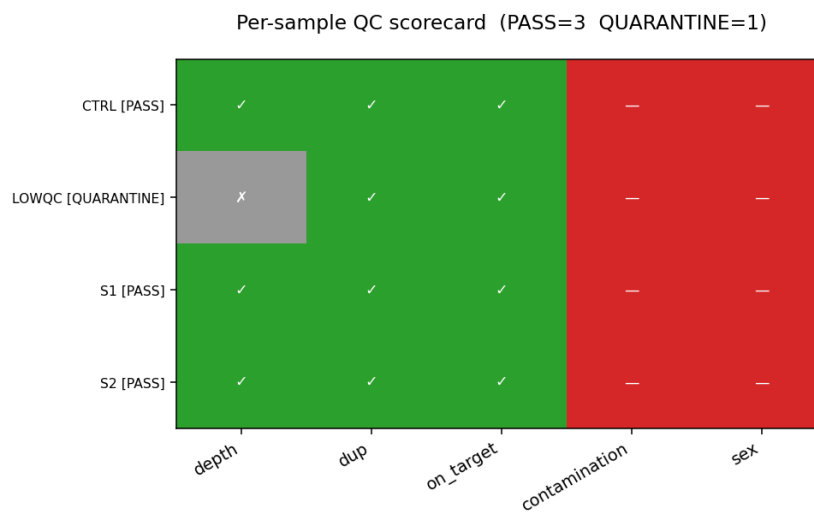


Figure 8: Per-sample QC checks (green=pass, red=fail, grey=not assessed); sample label shows the gate decision. Quarantined samples are kept and reported, not dropped.

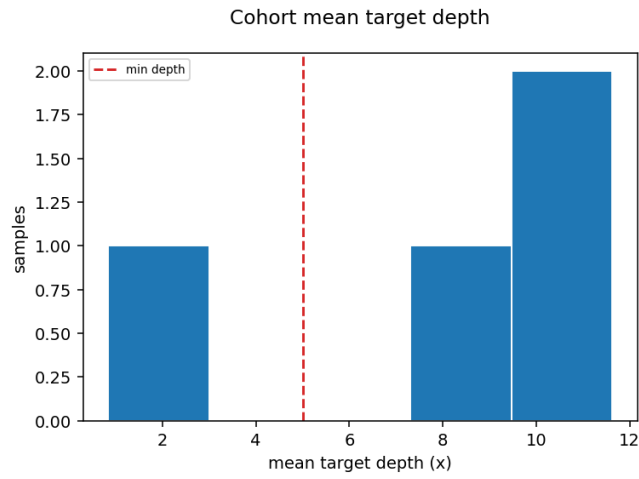


Figure 9: Distribution of per-sample mean target depth across the cohort vs the gate threshold.

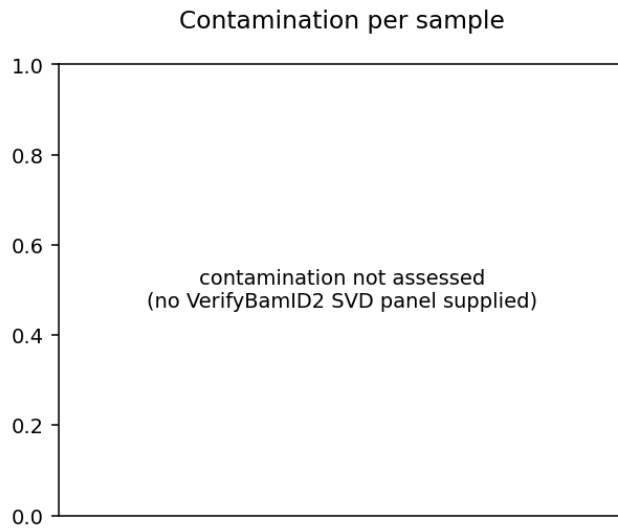


Figure 10: VerifyBamID2 FREEMIX contamination estimate per sample vs threshold (grey note if no SVD panel available).

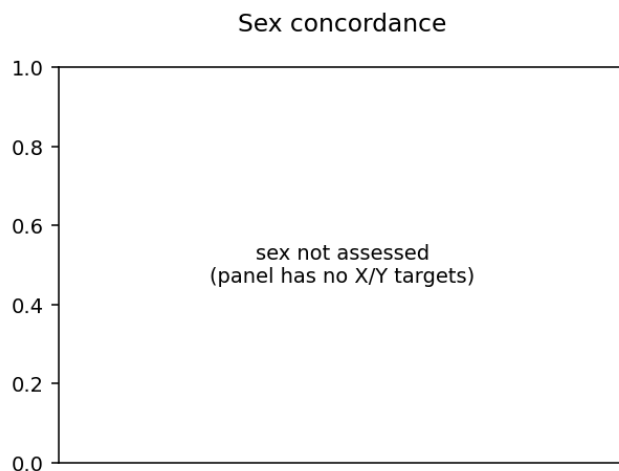


Figure 11: Declared vs inferred sex per sample (grey note if the panel has no X/Y targets to infer from).

9.3 Cohort QC dashboard & MultiQC (Stage 15)

`make_dashboard.py` builds `stage15_report/cohort_qc_dashboard.html` — a **self-contained** HTML embedding every captioned figure (base64) grouped by stage, plus the headline tables (resource manifest, QC scorecard, annotation landing, GIAB, burden). `multiqc_report.html` aggregates the per-tool outputs (FastQC, fastp, Picard, samtools, mosdepth).

10. Stage details & their plots

10.1 SNV/indel calling & filtering (Stages 6–7)

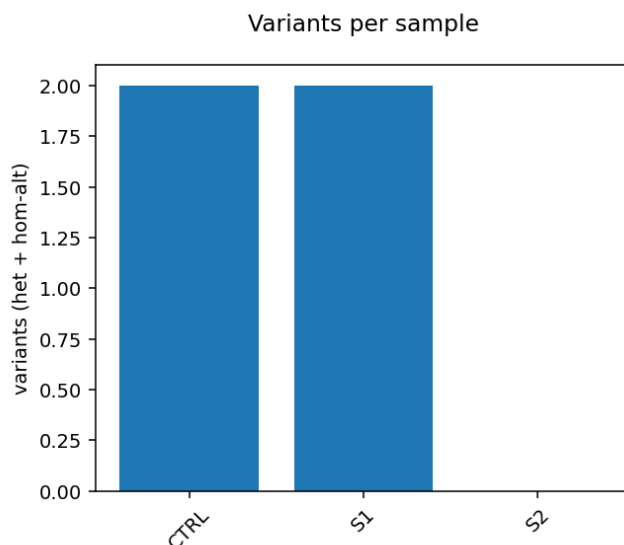


Figure 12: Per-sample count of non-reference genotypes in the joint callset (calling yield).

10.2 CNV, STR, paralog (Stages 8–10)

CNV calls (`cnv_calls.tsv`) carry a `cnv_callable` and `confidence` column from CaptureForge metadata — a call at a breakpoint-blind gene (CR1/CFH) is labelled `breakpoint_blind_low_confidence`, not silently trusted.

Paralog-aware flagging writes `INFO/PARALOG_GENE` and `INFO/PARALOG_CONF` into `paralog.annotated.vcf.gz` so each variant in a paralog region carries its confidence.

10.3 Annotation (Stage 11)

10.4 Cohort QC (Stage 12)

10.5 GIAB validation (Stage 13)

10.6 Burden (Stage 14)

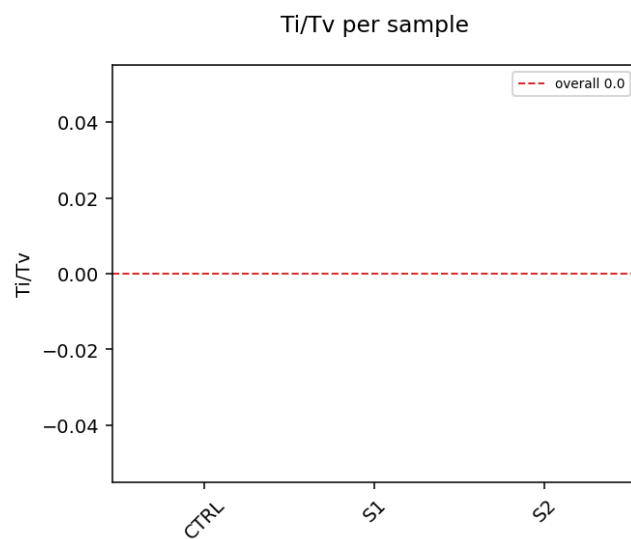


Figure 13: Transition/transversion ratio per sample; whole-exome ~ 3.0 , smaller panels vary (calling quality).

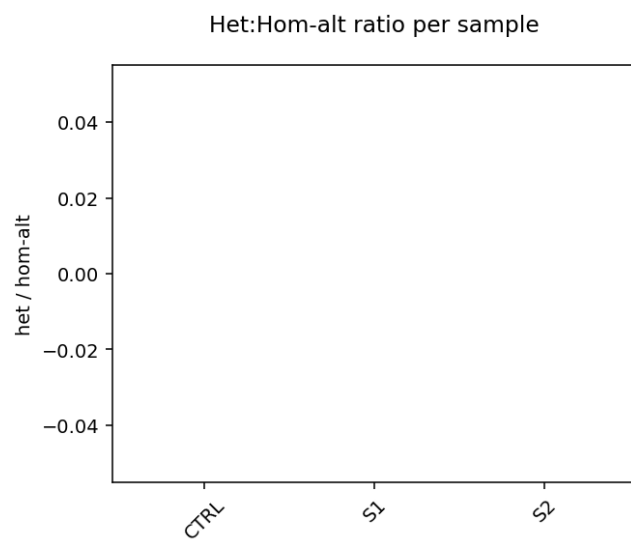


Figure 14: Heterozygous-to-homozygous-alt ratio per sample; extreme values flag sample-quality issues.

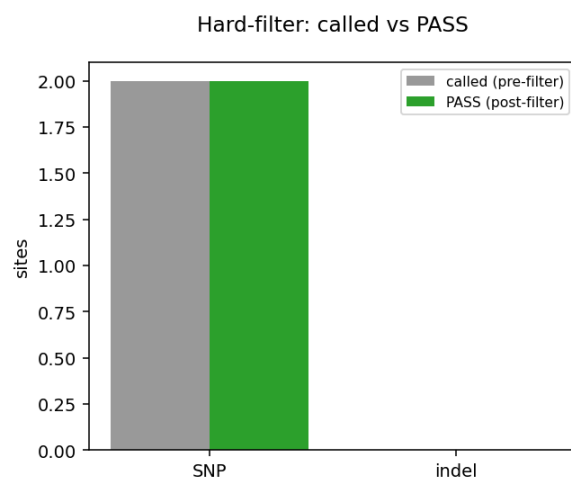


Figure 15: Variant counts before and after GATK hard-filtering, by type (sites retained).

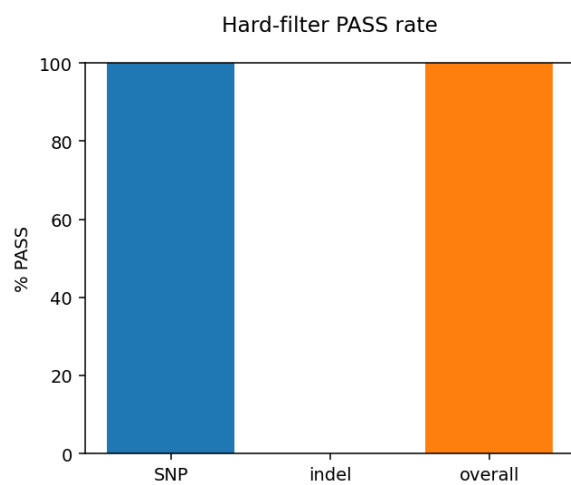


Figure 16: Fraction of called variants passing the GATK hard-filters, by type and overall.

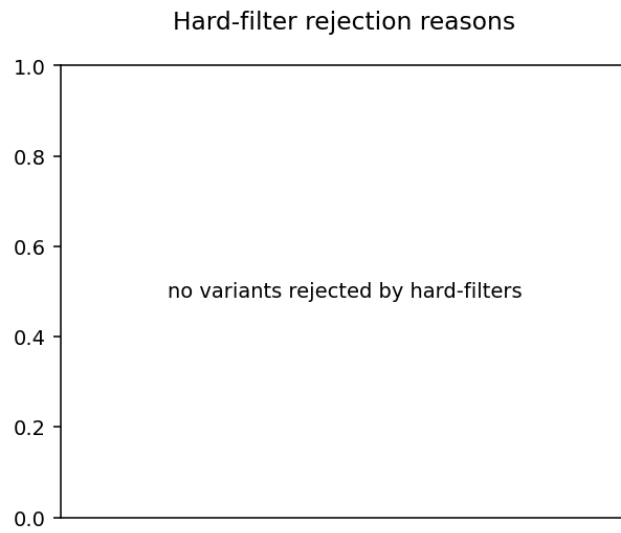


Figure 17: Number of sites rejected by each hard-filter criterion (which filter removed what).

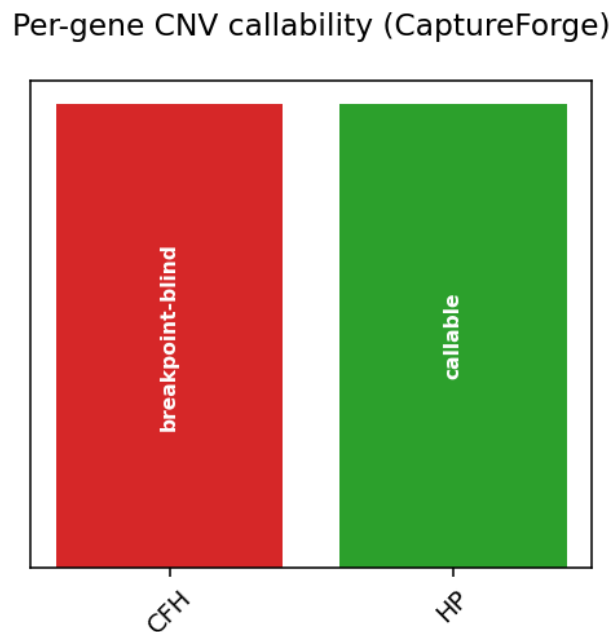


Figure 18: CNV-target genes coloured by CaptureForge callability: green=depth-callable, red=breakpoint-blind (low-confidence, e.g. CR1/CFH).

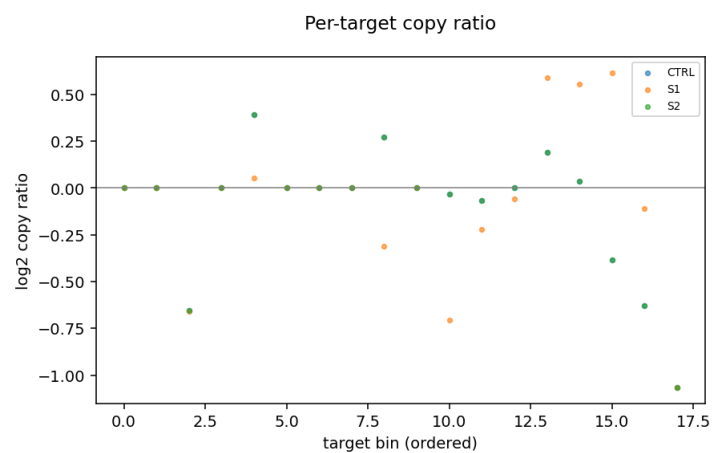


Figure 19: Per-target log2 copy ratio per sample; deviations from 0 indicate copy-number change.

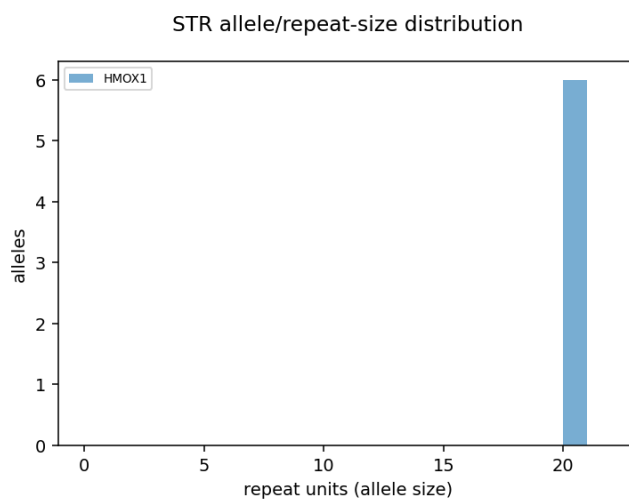


Figure 20: Distribution of genotyped repeat-unit counts (allele sizes) per STR locus.

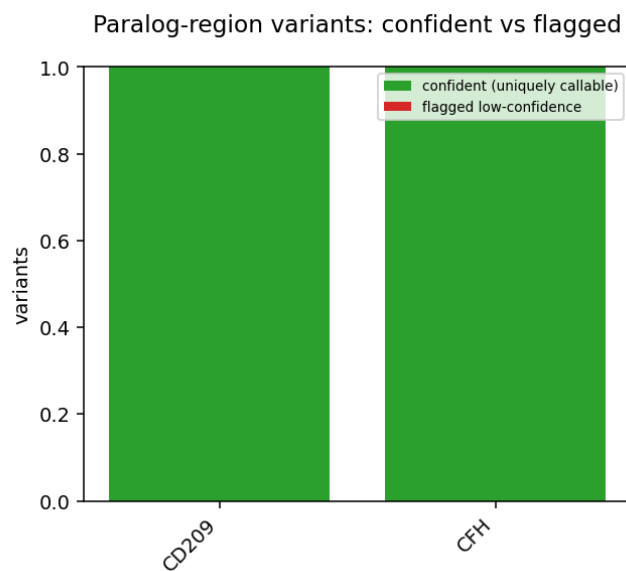


Figure 21: Variants in each paralog-ambiguous gene region, split into confident (uniquely callable, high MQ) vs flagged low-confidence (multimapping-ambiguous, low MQ).

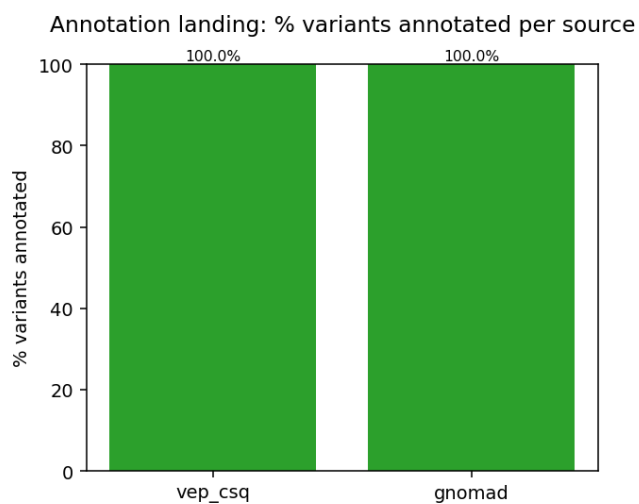


Figure 22: Fraction of callset variants that actually received each annotation (proof the annotation landed, not silently missed on a contig-name mismatch).

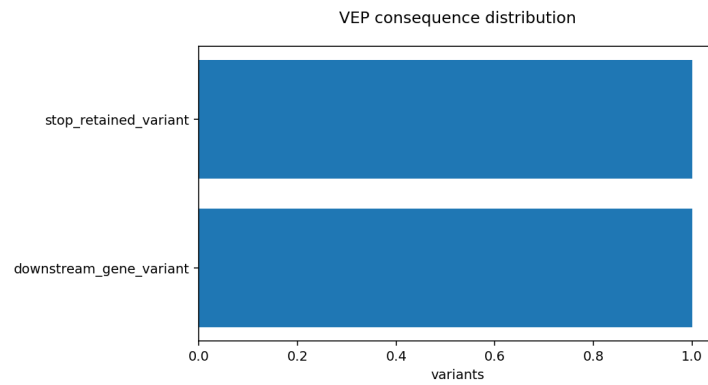


Figure 23: Most-severe VEP consequence per variant across the callset.

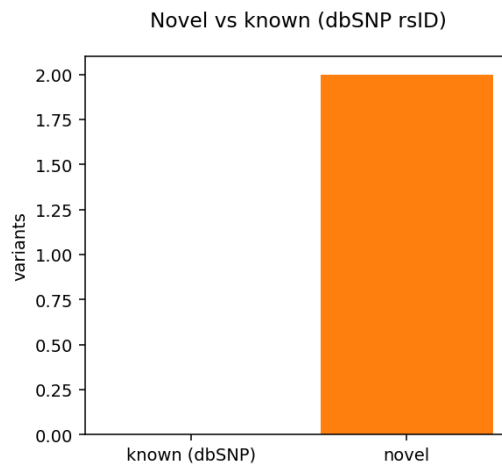


Figure 24: Variants with a dbSNP rsID (known) vs without (novel) — novelty rate.

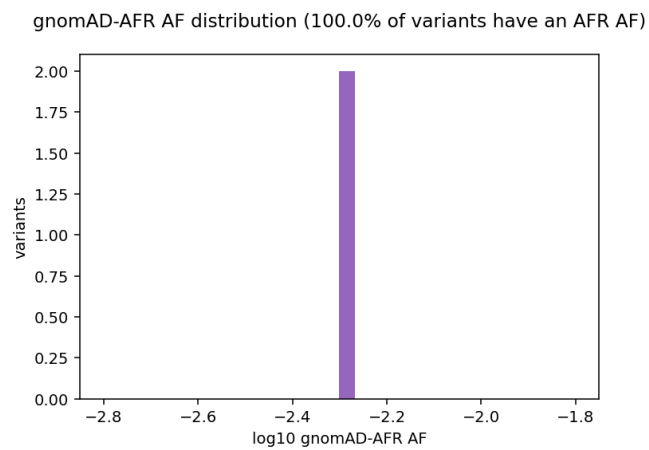


Figure 25: Distribution of gnomAD African-ancestry allele frequencies; title shows the % of variants carrying an AFR AF (key for rare-variant filtering / burden).

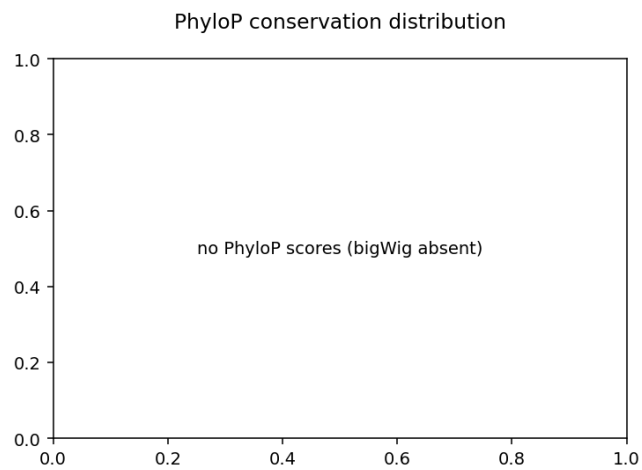


Figure 26: PhyloP conservation-score distribution; positive = conserved (likely functional).

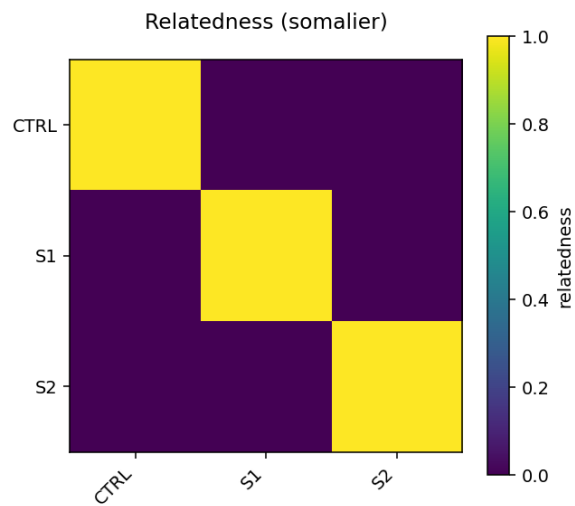


Figure 27: Pairwise somalier relatedness; synthetic test samples are near-identical so values run high (machinery proven) — a real cohort shows true relatedness structure.

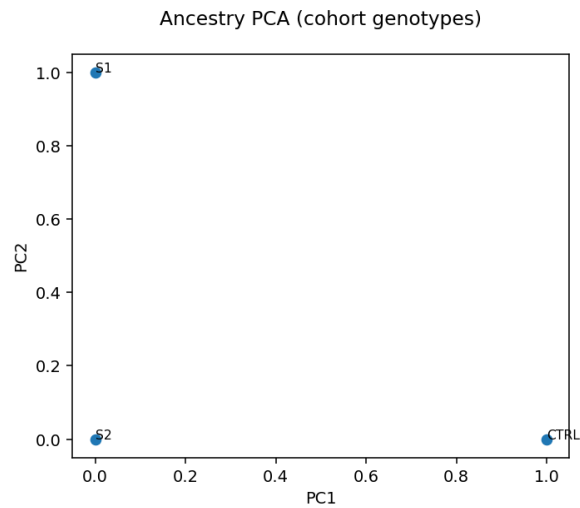


Figure 28: Genotype PCA of the cohort; on a tight panel few on-target sites give weak structure — real ancestry assignment uses somalier with the 1000G-labelled reference (off-target sites).

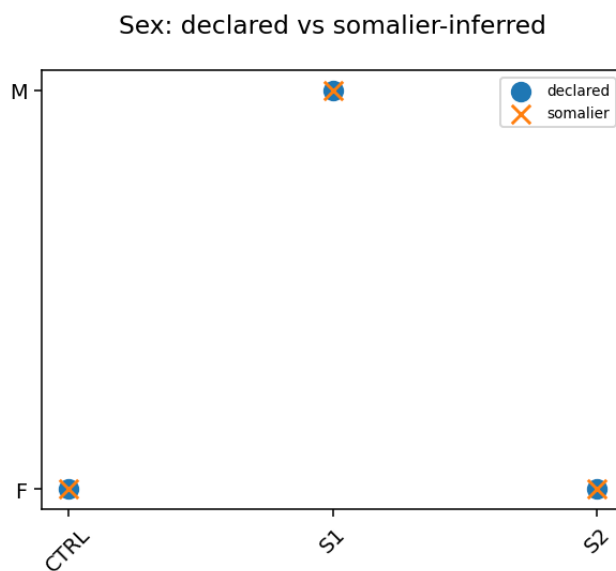


Figure 29: Declared vs somalier-inferred sex per sample (somalier infers from X/Y signal incl. off-target reads — the backstop for an autosomal panel with no X/Y targets).

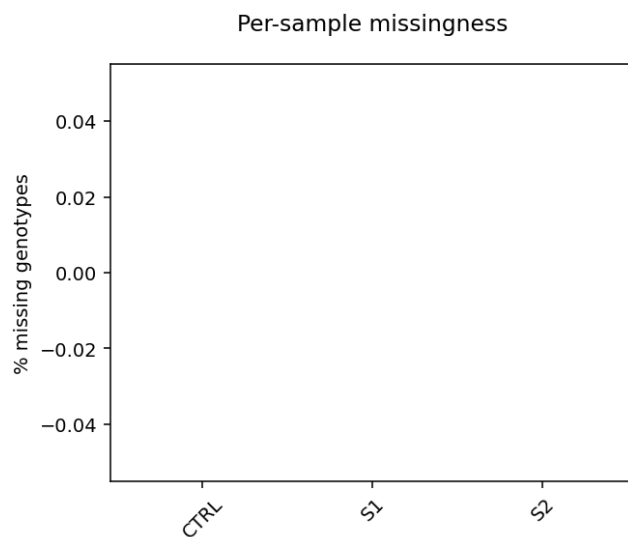


Figure 30: Per-sample fraction of no-call genotypes in the joint callset (sample quality).

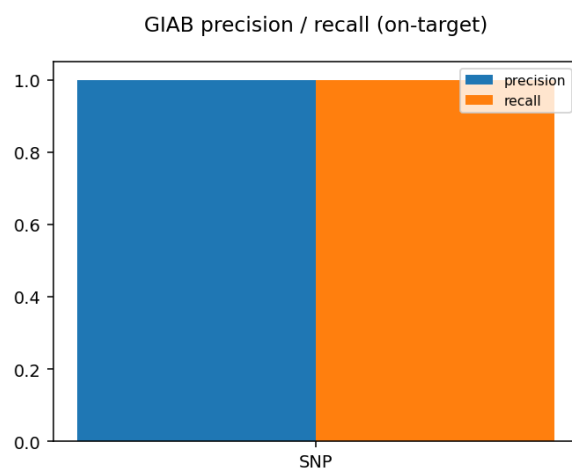


Figure 31: hap.py precision & recall vs GIAB truth, restricted to the panel BED (on-target). Synthetic test only proves the machinery; real numbers need a GIAB control + truth v4.2.1.

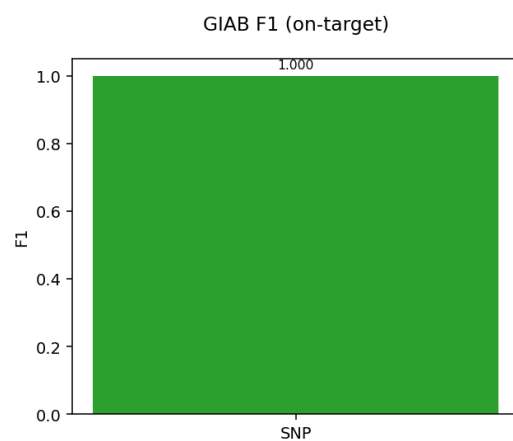


Figure 32: hap.py F1 per variant type vs GIAB truth, restricted to the panel BED (on-target).

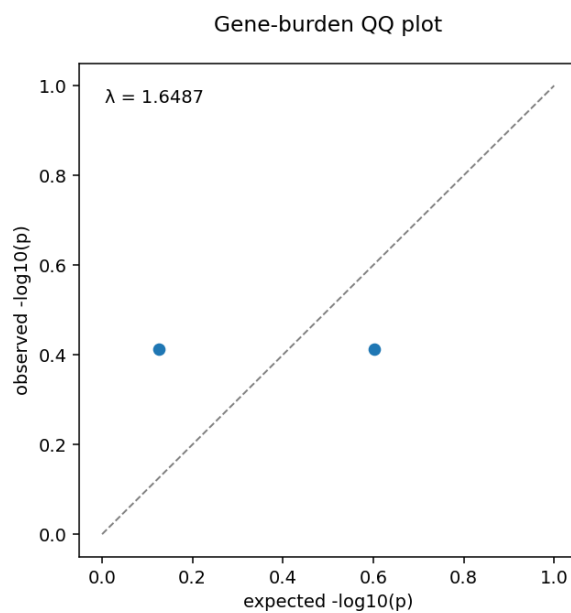


Figure 33: QQ plot of gene-burden p-values vs uniform expectation; lambda = genomic inflation. Small-N synthetic results prove the machinery only — not valid association.

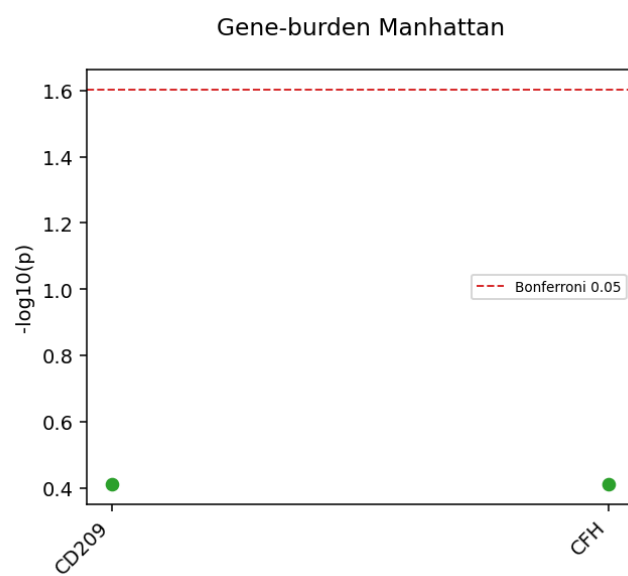


Figure 34: Per-gene burden significance ($-\log_{10} p$) with a Bonferroni line; collapses rare+functional variants per gene. Validity depends on N , phenotype definition, ancestry matching.

11. Outputs explained

Output	Location	File + notes
Annotated joint VCF	stage11_annotation/	annotated.vcf.gz — VEP CSQ + gnomAD_* + ClinVar_* + PhyloP + paralog flags
Per-variant TSV	stage11_annotation/	variants.flat.tsv — flattened (columns below)
Filtered joint VCF	stage7_filter/	joint.filtered.vcf.gz — FILTER = PASS or hard-filter name
CNV calls	stage8_cnv/	cnv_calls.tsv — with cnv_callable + confidence columns
STR genotypes	stage9_str/	str_calls.tsv — repeat units, read support, call rate
Paralog VCF	stage10_paralog/	paralog.annotated.vcf.gz — INFO/PARALOG_GENE, PARALOG_CONF
Burden results	stage14_burden/	burden_results.tsv — engine column stamps the method
Cohort QC	stage12_cohortqc/	cohort_qc.json — relatedness, sex, missingness, PCA
GIAB metrics	stage13_giab/	giab_metrics.json — precision/recall/F1, on-target
Provenance	stage15_report/	provenance.json — versions, resources, quarantine, PC gate
Dashboard	stage15_report/	cohort_qc_dashboard.html — all figures + tables

Per-variant TSV columns (variants.flat.tsv): chrom, pos, ref, alt, rsID, filter, gene, consequence, gnomAD_AF, gnomAD_AF_afr, ClinVar_CLNSIG, PhyloP, paralog_gene, paralog_conf.

Burden results columns (burden_results.tsv): unit, kind, engine, n_carriers, case_carriers, control_carriers, chi2, p (the engine column means a collapse result can never be mistaken for a regenie association when the file is read on its own).

12. Interpreting results

Machinery-proven vs needs-real-cohort. On the synthetic `test` cohort the pipeline *runs* and the *invariants* fire, but the *numbers* are not data quality. Treat as machinery-proven (not real): on-target/enrichment/uniformity, Ti/Tv & het:hom, GIAB precision/recall, all burden association (small N), and relatedness/ancestry structure.

CaptureForge confidence labels. Read CNV calls *through* their callability label: depth-callable genes (e.g. GYPC, HP) are trustworthy; breakpoint-blind genes (CR1, CFH) are `breakpoint_blind_low_confidence` — treat their CNV calls cautiously. Paralog-region variants are split **confident** (uniquely callable, high MQ) vs **flagged** (multimapping ambiguity) — confirm flagged calls orthogonally.

Burden validity. Small cohorts show **genomic inflation** (high λ in the QQ plot) — not signal. The **ancestry-PC gate** drops unstable PCs and runs **unadjusted** (stated in `provenance.json` and the burden summary); an unadjusted run is a deliberate, declared choice, not a silent one. Burden validity depends on N, phenotype definition, and ancestry matching — CallForge does not imply publication-ready association on a small cohort.

GIAB. Precision = fraction of called variants that are true; recall = fraction of truth recovered; both **on-target** (panel-BED-restricted). Genome-wide scoring would distort a panel’s measured sensitivity.

13. Real-run checklist

Before the production cohort:

- ☐ Demultiplexed FASTQs + sample sheet (with **phenotype** for burden).
- ☐ **Genome-wide gnomAD (AFR), ClinVar, dbSNP, PhyloP** — each must pass the scope gate. dbSNP is needed in **two** places (annotation rsIDs + BQSR known-sites).
- ☐ BQSR known-sites via `--known_sites` (dbSNP + Mills + 1000G indels).
- ☐ **1000G GRCh38 somalier sites** via `--somalier_sites`.
- ☐ GIAB control in the sheet + truth v4.2.1 (`--giab_control_id, --giab_truth_vcf/bed`).
- ☐ **Confirm ancestry PCs are stable** before using them as burden covariates (the gate enforces this; if unstable it runs unadjusted and says so).
- ☐ **Validate-before-production engines**: DeepVariant+GLnexus, regenie, SKAT-O/STAAR.
- ☐ Re-generate the genome-wide gnomAD panel slice on the HPC (fast network).
- ☐ Revisit GATK hard-filter thresholds for targeted data.

14. Troubleshooting

Symptom	Cause / fix
Scope-gate FAIL on a DB	region-subset DB — supply a genome-wide replacement (message names it)
Annotation values missing	chr-naming mismatch — handled by reconciliation; <code>verify_annotation.py</code> fails loud if not, check the manifest naming
VEP “CacheDir” error in gtf mode	do not pass <code>--offline</code> with <code>--gtf</code> (cache mode only)
mosdepth “exec format error” (arm64)	use mosdepth $\geq 0.3.11$ or <code>CONDA_SUBDIR=osx-64</code>
<code>grep -P</code> errors in fetch script (macOS)	BSD grep lacks <code>-P</code> ; the shipped script uses <code>awk</code> (keep it in sync)
No samples reach calling	all quarantined — read <code>qc_quarantine.txt</code> ; relax thresholds only with justification
Unstable ancestry PCs	the gate drops them and runs unadjusted (recorded); do not force them
Truncated <code>.bgz</code> DB	BGZF-EOF check flags it <code>truncated</code> — re-fetch (resumable)
Hard-filter over-removes on a panel	tune <code>--snp_filter_expr/--indel_filter_expr</code> (defaults are WGS/WES)

15. Extending CallForge

- **New annotation databases:** add them to a vcfanno TOML (`--vcfanno_toml`); the scope gate + landing check apply.
- **New organism:** `--species` + VEP cache + vcfanno TOML; human DBs skip (§6.6).
- **Engine choices:** §8.8 — note the validate-before-production engines.
- **Roadmap (not yet validated):** end-to-end runs of DeepVariant+GLnexus, regenie, and SKAT-O on a real cohort; VQSR for large cohorts; manifest-driven BQSR known-sites.

16. Reproducibility & provenance

Every run writes `stage15_report/provenance.json`: pipeline version + git commit, the reference + invariant statement, the resource manifest (status/build/md5/AFR), QC thresholds + pass/quarantine lists with reasons, the burden status + **ancestry-PC gate decision**, and pointers to the pinned `env/*.yaml` and the pinned VEP/hap.py containers. Pin Nextflow via `manifest.nextflowVersion` and tool versions via the env files; cite the exact reference and database releases recorded in the manifest.

17. Glossary & FAQ

gVCF — per-sample genomic VCF with reference-confidence blocks, the input to joint genotyping. **PoN** — panel-of-normals (CNVkit reference from QC-pass samples). **FREEMIX** — Verify-BamID2 contamination estimate. **FOLD_80** — uniformity penalty (1.0 = uniform). λ (**lambda**) — genomic inflation factor. **Burden test** — gene/gene-set aggregation of rare variants vs phenotype.

FAQ.

- *Do I need CaptureForge to run CallForge?* No — you can supply `--target_bed + --gene_metadata` directly; but the per-gene labels (CNV callability, paralog, burden groups) come from CaptureForge metadata and enrich the analysis.
- *What if I have no phenotype?* The pipeline runs through the annotated callset and skips burden with a message.
- *Why did a sample disappear from the callset?* It was quarantined — see `qc_quarantine.txt`.

18. Citation & the CaptureForge $\leftrightarrow$ CallForge pairing

Cite via `CITATION.cff` in the repository root. CallForge is the **analysis counterpart to CaptureForge**: CaptureForge designs the panel and records per-gene design knowledge; CallForge consumes that handoff and produces the analysed, annotated, QC'd callset and burden layer. Run them as a pair for a panel: design $\rightarrow$ sequence $\rightarrow$ **CallForge**.